

and take $M := \max \{f^s(k), 2k/\epsilon\}$; the second term in the maximum ensures that any $n \geq M$ is large enough to satisfy (5).

We finally have to show that every graph $G = (V, E)$ of order at least m has an ϵ -regular partition $\{V_0, V_1, \dots, V_{k'}\}$ with $m \leq k' \leq M$. So let G be given, and let $n := |G|$. If $n \leq M$, we partition G into $k' := n$ singletons, choosing $V_0 := \emptyset$ and $|V_1| = \dots = |V_{k'}| = 1$. This partition of G is clearly ϵ -regular. Suppose now that $n > M$. Let $C_0 \subseteq V$ be minimal such that k divides $|V \setminus C_0|$, and let $\{C_1, \dots, C_k\}$ be any partition of $V \setminus C_0$ into sets of equal size. Then $|C_0| \leq k - 1$, and hence $|C_0| \leq \epsilon n$ by (5). Starting with $\{C_0, C_1, \dots, C_k\}$ we apply Lemma 7.4.4 again and again, until the partition of G obtained is ϵ -regular; this will happen after at most s iterations, since by (5) the size of the exceptional set in the partitions stays below ϵn , so the lemma could indeed be reapplied up to the theoretical maximum of s times. $\square$

7.5 Applying the regularity lemma

The purpose of this section is to illustrate how the regularity lemma is typically applied in the context of (dense) extremal graph theory. Suppose we are trying to prove that a certain edge density of a graph G suffices to force the occurrence of some given subgraph H , and that we have an ϵ -regular partition of G . For most of the pairs (V_i, V_j) of partition sets, the edges between V_i and V_j are distributed fairly uniformly; their density, however, may depend on the pair. But since G has many edges, this density cannot be too small for too many pairs: some sizeable proportion of the pairs will have at least a certain positive density. Moreover if G is large, then so are the pairs: recall that the number of partition sets is bounded, and they have equal size. But any large enough bipartite graph with equal partition sets, fixed positive edge density (however small) and a uniform distribution of edges will contain any given bipartite subgraph;⁸ this will be made precise below. Writing H as a union of bipartite subgraphs, say those induced by pairs of colour classes of some vertex colouring of H , we shall obtain $H \subseteq G$ as desired.

These ideas will be formalized by the so-called ‘blow-up lemma’, Lemma 7.5.2 below. We shall then use this and the regularity lemma to prove the Erdős-Stone theorem from Section 7.1; another application will be given later, in the proof of Theorem 9.2.2. We wind up the section with an informal review of the application of the regularity lemma that we have seen, summarizing what it can teach us for similar applications. In particular, we look at how the various parameters involved depend

⁸ Readers already acquainted with random graphs may find it instructive to compare this statement with Proposition 11.3.1.

on each other, and in which order they have to be chosen to make the lemma work.

Let us begin by noting a simple consequence of the ϵ -regularity of a pair (A, B) . For any subset $Y \subseteq B$ that is not too small, most vertices of A have about the expected number of neighbours in Y :

Lemma 7.5.1. *Let (A, B) be an ϵ -regular pair, of density d say, and let $Y \subseteq B$ have size $|Y| \geq \epsilon|B|$. Then all but fewer than $\epsilon|A|$ of the vertices in A have (each) at least $(d - \epsilon)|Y|$ neighbours in Y .*

Proof. Let $X \subseteq A$ be the set of vertices with fewer than $(d - \epsilon)|Y|$ neighbours in Y . Then $\|X, Y\| < |X|(d - \epsilon)|Y|$, so

$$d(X, Y) = \frac{\|X, Y\|}{|X||Y|} < d - \epsilon = d(A, B) - \epsilon.$$

As (A, B) is ϵ -regular and $|Y| \geq \epsilon|B|$, this implies that $|X| < \epsilon|A|$. $\square$

Let G be a graph with an ϵ -regular partition $\{V_0, V_1, \dots, V_k\}$, with exceptional set V_0 and $|V_1| = \dots = |V_k| =: \ell$. Given $d \in [0, 1]$, let R be the graph on $\{1, \dots, k\}$ in which two vertices i, j are adjacent if and only if (V_i, V_j) is ϵ -regular of density $\geq d$. We shall call R a *regularity graph* of G with parameters ϵ, ℓ and d . Given $s \in \mathbb{N}$, let us now replace every vertex i of R by a set V_i^s of s vertices, and every edge by a complete bipartite graph between the corresponding s -sets. The resulting graph will be denoted by R_s . (For $R = K^r$, for example, we have $R_s = K_s^r$.)

R
regularity
graph
 V_i^s
 R_s

Lemma 7.5.2, a simple version of the blow-up lemma that suffices for our intended applications of the regularity lemma, says that subgraphs of R_s can also be found in G , provided that $d > 0$, that ϵ is small enough, and that the V_i are large enough. In fact, the values of ϵ and ℓ required depend only on $(d \text{ and})$ the maximum degree of the subgraph:

Lemma 7.5.2. (Blow-up Lemma⁹)

For all $d \in (0, 1]$ and $\Delta \geq 1$ there exists an $\epsilon_0 > 0$ with the following property: if G is any graph, H is a graph with $\Delta(H) \leq \Delta$, $s \in \mathbb{N}$, and R is any regularity graph of G with parameters $\epsilon \leq \epsilon_0$, $\ell \geq 2s/d^\Delta$ and d , then

[9.2.2]

$$H \subseteq R_s \Rightarrow H \subseteq G.$$

Proof. Given d and Δ , choose $\epsilon_0 > 0$ small enough that $\epsilon_0 < d$ and

d, Δ, ϵ_0

$$(d - \epsilon_0)^\Delta - \Delta\epsilon_0 \geq \frac{1}{2}d^\Delta; \quad (1)$$

such a choice is possible, since $(d - \epsilon)^\Delta - \Delta\epsilon \rightarrow d^\Delta$ as $\epsilon \rightarrow 0$. Now let G, H, R, R_s

⁹ See the notes for the original, more general, blow-up lemma.

V_i G, H, s and R be given as stated. Let $\{V_0, V_1, \dots, V_k\}$ be the ϵ -regular partition of G that gave rise to R ; thus, $\epsilon \leq \epsilon_0$ and $V(R) = \{1, \dots, k\}$ and $|V_1| = \dots = |V_k| = \ell \geq 2s/d^\Delta$. Let us assume that H is actually a subgraph of R_s (not just isomorphic to one), with vertices $u_1, \dots, u_h$ say. Each vertex u_i lies in one of the s -sets V_j^s of R_s , which defines a map $\sigma: i \mapsto j$. Our aim is to define an embedding $u_i \mapsto v_i \in V_{\sigma(i)}$ of H in G as a subgraph; thus, $v_1, \dots, v_h$ will be distinct, and $v_i v_j$ will be an edge of G whenever $u_i u_j$ is an edge of H .

ϵ, k, ℓ
 u_i, h

σ
 v_i

Our plan is to choose the vertices $v_1, \dots, v_h$ inductively. Throughout the induction, we shall have a ‘target set’ $Y_i \subseteq V_{\sigma(i)}$ assigned to each u_i ; this contains the vertices that are still candidates for the choice of v_i . Initially, Y_i is the entire set $V_{\sigma(i)}$. As the embedding proceeds, Y_i will get smaller and smaller (until it collapses to $\{v_i\}$ when v_i is chosen): whenever we choose a vertex v_j with $j < i$ and $u_j u_i \in E(H)$, we delete all those vertices from Y_i that are not adjacent to v_j . The set Y_i thus evolves as

$$V_{\sigma(i)} = Y_i^0 \supseteq \dots \supseteq Y_i^i = \{v_i\},$$

where Y_i^j denotes the version of Y_i current after the definition of v_j and the resulting deletion of vertices from Y_i^{j-1} .

In order to make this approach work, we have to ensure that the target sets Y_i do not get too small. When we come to embed a vertex u_j , we consider all the indices $i > j$ with $u_j u_i \in E(H)$; there are at most Δ such i . For each of these i , we wish to select v_j so that

$$Y_i^j = N(v_j) \cap Y_i^{j-1} \quad (2)$$

is still relatively large: smaller than Y_i^{j-1} by no more than a constant factor such as $(d - \epsilon)$. Now this can be done by Lemma 7.5.1 (with $A = V_{\sigma(j)}$, $B = V_{\sigma(i)}$ and $Y = Y_i^{j-1}$): provided that Y_i^{j-1} still has size at least $\epsilon\ell$ (which induction will ensure), all but at most $\epsilon\ell$ choices of v_j will be such that the new set Y_i^j as in (2) satisfies

$$|Y_i^j| \geq (d - \epsilon)|Y_i^{j-1}|. \quad (3)$$

Excluding the bad choices for v_j for all the relevant values of i simultaneously, we find that all but at most $\Delta\epsilon\ell$ choices of v_j from $V_{\sigma(j)}$, and in particular from $Y_j^{j-1} \subseteq V_{\sigma(j)}$, satisfy (3) for all i .

It remains to show that the sets Y_i^{j-1} considered above as Y for Lemma 7.5.1 never fall below the size of $\epsilon\ell$, and that when we come to select $v_j \in Y_j^{j-1}$ we have a choice of at least s suitable candidates: since before u_j at most $s - 1$ vertices u were given an image in $V_{\sigma(j)}$, we can then choose v_j distinct from these.

But all this follows from our choice of ϵ_0 . Indeed, the initial target sets Y_i^0 have size ℓ , and each Y_i shrinks at most Δ times by a factor of $(d - \epsilon)$ when some v_j with $j < i$ and $u_j u_i \in E(H)$ is defined. Thus,

$$|Y_i^{j-1}| - \Delta\epsilon\ell \underset{(3)}{\geq} (d - \epsilon)^\Delta \ell - \Delta\epsilon\ell \geq (d - \epsilon_0)^\Delta \ell - \Delta\epsilon_0\ell \underset{(1)}{\geq} \frac{1}{2}d^\Delta \ell \geq s$$

for all $j \leq i$; in particular, we have $|Y_i^{j-1}| \geq \epsilon\ell$ and $|Y_j^{j-1}| - \Delta\epsilon\ell \geq s$ as desired. $\square$

We are now ready to prove the Erdős-Stone theorem.

Proof of Theorem 7.1.2. Let $r \geq 2$ and $s \geq 1$ be given as in the statement of the theorem. For $s = 1$ the assertion follows from Turán's theorem, so we assume that $s \geq 2$. Let $\gamma > 0$ be given; this γ will play the role of the ϵ of the theorem. If any graph G with $|G| =: n$ has

$$\|G\| \geq t_{r-1}(n) + \gamma n^2 \tag{7.1.1} \tag{7.1.4} \tag{7.4.1} \tag{r, s} \tag{\gamma} \tag{\|G\|}$$

edges, then $\gamma < 1$. We want to show that $K_s^r \subseteq G$ if n is large enough.

Our plan is to use the regularity lemma to show that G has a regularity graph R dense enough to contain a K^r by Turán's theorem. Then R_s contains a K_s^r , so we may hope to use our blow-up lemma, Lemma 7.5.2, to deduce that $K_s^r \subseteq G$.

On input $d := \gamma$ and $\Delta := \Delta(K_s^r)$ Lemma 7.5.2 returns an $\epsilon_0 > 0$. To apply the regularity lemma, let $m > 1/\gamma$ and choose $\epsilon > 0$ small enough that $\epsilon \leq \epsilon_0$,

$$\epsilon < \gamma/2 < 1, \tag{1}$$

and

$$\delta := 2\gamma - \epsilon^2 - 4\epsilon - d - \frac{1}{m} > 0; \tag{\delta}$$

this is possible, since $2\gamma - d - \frac{1}{m} > 0$. On input ϵ and m , the regularity lemma returns an integer M . Let us assume that

$$n \geq \frac{2Ms}{d^\Delta(1 - \epsilon)}. \tag{n}$$

Since this number is at least m , the regularity lemma provides us with an ϵ -regular partition $\{V_0, V_1, \dots, V_k\}$ of G , where $m \leq k \leq M$; let $|V_1| = \dots = |V_k| =: \ell$. Then

$$n \geq k\ell, \tag{2}$$

and

$$\ell = \frac{n - |V_0|}{k} \geq \frac{n - \epsilon n}{M} = n \frac{1 - \epsilon}{M} \geq \frac{2s}{d^\Delta}$$

R

by the choice of n . Let R be the regularity graph of G with parameters ϵ, ℓ, d corresponding to the above partition. Then Lemma 7.5.2 will imply $K_s^r \subseteq G$ as desired if $K^r \subseteq R$ (and hence $K_s^r \subseteq R_s$).

Our plan was to show $K^r \subseteq R$ by Turán's theorem. We thus have to check that R has enough edges, i.e. that enough ϵ -regular pairs (V_i, V_j) have density at least d . This should follow from our assumption that G has at least $t_{r-1}(n) + \gamma n^2$ edges, i.e. an edge density of about $\frac{r-2}{r-1} + 2\gamma$: this lies substantially above the approximate density of $\frac{r-2}{r-1}$ of the Turán graph $T^{r-1}(k)$, and hence substantially above any density that G could derive from $t_{r-1}(k)$ dense pairs alone, even if all these had density 1.

Let us then estimate $\|R\|$ more precisely. How many edges of G lie outside ϵ -regular pairs? At most $\binom{|V_0|}{2}$ edges lie inside V_0 , and by condition (i) in the definition of ϵ -regularity these are at most $\frac{1}{2}(\epsilon n)^2$ edges. At most $|V_0|k\ell \leq \epsilon n^2$ edges join V_0 to other partition sets. The at most ϵk^2 other pairs (V_i, V_j) that are not ϵ -regular contain at most ℓ^2 edges each, together at most $\epsilon k^2 \ell^2$. The ϵ -regular pairs of insufficient density ($< d$) each contain no more than $d\ell^2$ edges, altogether at most $\frac{1}{2}k^2 d\ell^2$ edges. Finally, there are at most $\binom{\ell}{2}$ edges inside each of the partition sets $V_1, \dots, V_k$, together at most $\frac{1}{2}\ell^2 k$ edges. All *other* edges of G lie in ϵ -regular pairs of density at least d , and thus contribute to edges of R . Since each edge of R corresponds to at most ℓ^2 edges of G , we thus have in total

$$\|G\| \leq \frac{1}{2}\epsilon^2 n^2 + \epsilon n^2 + \epsilon k^2 \ell^2 + \frac{1}{2}k^2 d\ell^2 + \frac{1}{2}\ell^2 k + \|R\| \ell^2.$$

Hence, for all sufficiently large n ,

$$\begin{aligned} \|R\| &\geq \frac{1}{2}k^2 \frac{\|G\| - \frac{1}{2}\epsilon^2 n^2 - \epsilon n^2 - \epsilon k^2 \ell^2 - \frac{1}{2}dk^2 \ell^2 - \frac{1}{2}k\ell^2}{\frac{1}{2}k^2 \ell^2} \\ &\stackrel{(1,2)}{\geq} \frac{1}{2}k^2 \left(\frac{t_{r-1}(n) + \gamma n^2 - \frac{1}{2}\epsilon^2 n^2 - \epsilon n^2}{n^2/2} - 2\epsilon - d - \frac{1}{k} \right) \\ &\geq \frac{1}{2}k^2 \left(\frac{t_{r-1}(n)}{n^2/2} + 2\gamma - \epsilon^2 - 4\epsilon - d - \frac{1}{m} \right) \\ &= \frac{1}{2}k^2 \left(t_{r-1}(n) \binom{n}{2}^{-1} \left(1 - \frac{1}{n} \right) + \delta \right) \\ &> \frac{1}{2}k^2 \frac{r-2}{r-1} \\ &\geq t_{r-1}(k). \end{aligned}$$

(The strict inequality follows from Lemma 7.1.4.) Therefore $K^r \subseteq R$ by Theorem 7.1.1, as desired. $\square$

Having seen a typical application of the regularity lemma in full detail, let us now step back and try to separate the wheat from the chaff: what were the main ideas, how do the various parameters depend on each other, and in which order were they chosen?

The task was to show that γn^2 more edges than can be accommodated on n vertices without creating a K^r force a K_s^r subgraph, provided that G is large enough. The plan was to do this using our blow-up lemma, which asks for the input of two parameters: d and Δ . As we wish to find a copy of $H = K_s^r$ in G , it is clear that we must choose $\Delta := \Delta(K_s^r)$. We shall return to the question of how to choose d in a moment.

Given d and Δ , our blow-up lemma tells us how small we must choose ϵ to make the regularity lemma provide us with a suitable partition. The regularity lemma also requires the input of a lower bound m for the number of partition classes; we shall discuss this below, together with d .

All that remains now is to choose G large enough that the partition classes have size at least $2s/d^\Delta$, as required by the blow-up lemma. (The s here depends on the graph H we wish to embed, and $s := |H|$ would certainly be big enough. In our case, we can use the s from our $H = K_s^r$.) How large is ‘large enough’ for $|G|$ follows straight from the upper bound M on the number of partition classes returned by the regularity lemma: roughly, i.e. disregarding V_0 , an assumption of $|G| \geq 2Ms/d^\Delta$ suffices.

So far, everything was entirely straightforward, and standard for any application of the regularity lemma of this kind. But now comes the interesting bit, the part specific to this proof: the observation that, if only d is small enough, our γn^2 ‘additional edges’ force an ‘additional dense ϵ -regular pair’ of partition sets, giving us more than $t_{r-1}(k)$ dense ϵ -regular pairs in total (where ‘dense’ means ‘of density at least d ’), thus forcing R to contain a K^r and hence R_s to contain a K_s^r .

Let us examine why this is so. Suppose we have at most $t_{r-1}(k)$ dense ϵ -regular pairs. Inside these, G has at most

$$\frac{1}{2}k^2 \frac{r-2}{r-1} \ell^2 \leq \frac{1}{2}n^2 \frac{r-2}{r-1}$$

edges, even if we use those pairs to their full capacity of ℓ^2 edges each (where ℓ is again the common size of the partition sets other than V_0 , so that $k\ell$ is nearly n). Thus, we have almost exactly our γn^2 additional edges left to accommodate elsewhere in the graph: either in ϵ -regular pairs of density less than d , or in some exceptional way, i.e. in irregular pairs, inside a partition set, or with an end in V_0 . Now the number of edges in low-density ϵ -regular pairs is less than

$$\frac{1}{2}k^2 d \ell^2 \leq \frac{1}{2}dn^2,$$

and hence less than half of our extra edges if $d \leq \gamma$. The other half, the remaining $\frac{1}{2}\gamma n^2$ edges, are more than can be accommodated in ex-

ceptional ways, provided we choose m large enough and ϵ small enough (giving an additional upper bound for ϵ). It is now a routine matter to compute the values of m and ϵ that will work.

7.6 Two regularity tools

Our acquaintance with the regularity lemma so far already includes the essentials: the notion of regularity, a statement of the regularity lemma, its proof, and a typical application of the lemma in our proof of Theorem 7.1.2 in Section 7.5. Another application, also based on no more than we have covered so far, will be given in the proof of Theorem 9.2.2. Both these applications illustrate techniques typically involved in many uses of the regularity lemma.

Our aim in this section is to extract some of these techniques and bundle them into formal tools that can be used out of the box: the ‘counting lemma’, the ‘removal lemma’, and some immediate consequences of these. Their proofs will not be short, but the ideas they involve will look familiar by now. Conversely, our proofs of Theorems 7.1.2 and 9.2.2 could be rewritten to use some of these formal tools (Exercise 45).

To facilitate the proof of the counting lemma, it will be convenient to use a slightly different notion of ϵ -regularity. To avoid confusion, we shall call this ‘ ϵ -uniformity’ here; but in the literature it is usually referred to as (another version of) ‘ ϵ -regularity’ as well.

Call a pair (A, B) of disjoint sets of vertices in a graph (ϵ, d) -uniform for $\epsilon > 0$ and $0 \leq d \leq 1$ if their non-empty subsets $X \subseteq A$ and $Y \subseteq B$ satisfy

$$|\|X, Y\| - d|X||Y|| \leq \epsilon|A||B|,$$

or equivalently,

$$|d(X, Y) - d| \leq \epsilon \frac{|A||B|}{|X||Y|}.$$

We may think of the real number d as a ‘target density’ with which the densities $d(X, Y) = \|X, Y\|/|X||Y|$ of the pairs (X, Y) are compared. If this is chosen as $d = d(A, B)$, the (ϵ, d) -uniform pair (A, B) is simply called ϵ -uniform.

This differs from our definition of ϵ -regularity in Section 7.4 in two ways. One is that all subsets $X \subseteq A$ and $Y \subseteq B$ are required to satisfy the inequality, including those of size less than $\epsilon|A|$ or $\epsilon|B|$. The other difference is best seen in the second of the two inequalities. Its left-hand side measures how much the density of (X, Y) differs from that of (A, B) . Our definition of ϵ -regularity requires that this should be at most ϵ . This is relaxed here by the factor $|A||B|/|X||Y| \geq 1$: the density of small pairs (X, Y) is allowed to deviate from d more than that of bigger ones.

(ϵ, d) -
uniform

ϵ -uniform